ProtoGen: Federated Privacy-Preserving Prototype Learning via Feature Generation in Heterogeneous Environments (Supplementary Resource)

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I. Theoretical Analysis of ProtoGen

This section presents the following theorems, which collectively demonstrate the theoretical foundations of ProtoGen.

A. Preliminaries and Notation

Let $\mathbb{K}$ be the set of all clients, with $|\mathbb{K}| = K$. For each client $k \in \mathbb{K}$, let $\mathbb{D}_k$ be its local dataset, with $n_k = |\mathbb{D}_k|$ and $n = \sum_{k=1}^{K} n_k$. Let $\theta_{k,t} \in \mathbb{R}^d$ denote the local model parameters of client k at round t. Let $\rho_{k,t} \in \mathbb{R}^p$ and $\varrho_{k,t} \in \mathbb{R}^q$ denote the *PPGM* encoder and decoder parameters of client k at round t, respectively. $\psi_k \leftarrow \{\rho_k, \varrho_k\}$ represents the general *PPGM* model parameters of client k.

We define the following:

- $\mathcal{L}_C(\theta; x, y)$: loss function for client models
- $\mathcal{L}_{GEN}(\psi; x, y)$: loss function for *PPGM*
- $\mathcal{T}(\theta_k) = \mathbb{E}_{(x,y)\sim\mathbb{D}_k}[\mathcal{L}_C(\theta_k; x, y)]$: expected combined local objective loss for client k
- $\mathcal{V}(\psi_k) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{(x,y)\sim\mathbb{D}_k}[\mathcal{L}_{GEN}(\psi_k; x, y)]$: expected feature generation loss for client k.

B. Assumptions

We make the following assumptions:

Assumption 1 (Smoothness): $\mathcal{T}(\theta_k)$ is L-smooth for all k: $\forall \theta_k, \theta'_k \in \mathbb{R}^d$,

$$\|\nabla\mathcal{T}(\theta_k) - \nabla\mathcal{T}(\theta'_k)\| \leq L_1\|\theta_k - \theta'_k\| \tag{1}$$

where $L_1 > 0$ is the Lipschitz constant for the local model.

Assumption 2 (Strong Convexity): $\mathcal{T}(\theta_k)$ is μ-strongly convex for all k: $\forall \theta_k, \theta'_k \in \mathbb{R}^d$,

$$\mathcal{T}(\theta'_k) \geq \mathcal{T}(\theta_k) + \langle\nabla\mathcal{T}(\theta_k), \theta'_k - \theta_k\rangle + \frac{\mu}{2}\|\theta'_k - \theta_k\|^2 \tag{2}$$

Assumption 3 (Bounded Variance): The variance of stochastic gradients is bounded for both client models and *PPGM*, $\forall k \in \mathbb{K}, \forall \theta_k \in \mathbb{R}^d$:

$$\mathbb{E}_{(x,y)\sim\mathbb{D}_k}[\|\nabla\mathcal{L}_C(\theta_k; x, y) - \nabla\mathcal{T}(\theta_k)\|^2] \leq \sigma_1^2 \tag{3}$$

$$\mathbb{E}_{(x,y)\sim\mathbb{D}_k}[\|\nabla\mathcal{L}_{GEN}(\psi_k; x, y) - \nabla\mathcal{V}(\psi_k)\|^2] \leq \sigma_2^2 \tag{4}$$

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*Assumption 4 (*PPGM *Lipschitz Continuity):* The *PPGM* model is Lipschitz continuous with respect to its parameters:

$$\|\nabla\mathcal{V}(\psi_k) - \nabla\mathcal{V}(\psi'_k)\| \leq L_2(\|\psi_k - \psi'_k\|) \tag{5}$$

where $L_2 > 0$ is the Lipschitz constant for the *PPGM* model.

Assumption 5 (Gradient Dissimilarity): For each client $k \in \mathbb{K}$, assume that the loss function $\mathcal{T}(\theta_k)$ is differentiable and bounded below, with $\theta_{k,t}$ denoting the model parameters at round t, and θ_k^* being a minimizer of $\mathcal{T}(\theta_k)$. Then, there exist constants $G \geq 0$ and $\beta \geq 0$ such that:

$$\frac{1}{K}\sum_{k=1}^{K}\|\nabla\mathcal{T}(\theta_{k,0}) - \nabla\mathcal{T}(\theta_k^*)\|^2 \leq G^2 + 2\beta\left(\mathcal{T}(\theta_{k,0}) - \mathcal{T}(\theta_k^*)\right) \tag{6}$$

where G^2 is a non-negative constant that upper bounds the inherent variance in the client gradients (even near local optima), β is a non-negative constant that quantifies how gradient differences grow with suboptimality (i.e., how far $\theta_{k,t}$ is from θ_k^* in terms of objective value).

C. Convergence Analysis

Lemma 1 (One-step Progress for Client Models): For client $k \in \mathbb{K}$, gradient computed with learning rate η_l over E local epochs at round $t \geq 0$, its one-step convergence bound follows:

$$\begin{aligned}\mathbb{E}[\|\theta_{k,t+1} - \theta_k^*\|^2] &\leq (1 - \mu\eta_l)\mathbb{E}[\|\theta_{k,t} - \theta_k^*\|^2] \\ &\quad - 2\eta_l\mathbb{E}[\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)] \\ &\quad + \eta_l^2\left(\sigma_1^2 + 6L\gamma_1 E\eta_l + 8(E-1)^2\gamma_1^2\right)\end{aligned} \tag{7}$$

where $\gamma_1 = \max\{\frac{8L_1}{\mu}, E\}$

Proof 1: Let $g_{k,t}$ be the average stochastic gradient computed by client k at round t over E local epochs. The update rule for client k's model parameters is $\theta_{k,t+1} = \theta_{k,t} - \eta_l g_{k,t}$. We begin by expanding $\|\theta_{k,t+1} - \theta_k^*\|^2$:

$$\begin{aligned}\|\theta_{k,t+1} - \theta_k^*\|^2 &= \|\theta_{k,t} - \eta_l g_{k,t} - \theta_k^*\|^2 \\ &= \|\theta_{k,t} - \theta_k^*\|^2 - 2\eta_l\langle g_{k,t}, \theta_{k,t} - \theta_k^*\rangle \\ &\quad + \eta_l^2\|g_{k,t}\|^2\end{aligned}$$

Taking the expectation of both sides:

$$\begin{aligned}\mathbb{E}[\|\theta_{k,t+1} - \theta_k^*\|^2] = \mathbb{E}[\|\theta_{k,t} - \theta_k^*\|^2] - 2\eta_l\mathbb{E}[\langle g_{k,t}, \theta_{k,t} - \theta_k^*\rangle] \\ + \eta_l^2\mathbb{E}[\|g_{k,t}\|^2]\end{aligned}$$

Using the strong convexity and smoothness of $T(\theta_k)$, we can bound the inner product term:

$$\langle g_{k,t}, \theta_{k,t} - \theta_k^* \rangle \geq \mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*) + \frac{\mu}{2}\|\theta_{k,t} - \theta_k^*\|^2 - \langle g_{k,t} - \nabla\mathcal{T}(\theta_{k,t}), \theta_{k,t} - \theta_k^* \rangle$$

For the gradient norm, we use the smoothness property and the definition of γ_1:

$$\mathbb{E}[\|g_{k,t}\|^2] \leq 4L_1(\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)) + 2\sigma_1^2 + 2\gamma_1^2 \quad (8)$$

Combining these bounds and applying Cauchy-Schwarz inequality to the error term, we get:

$$\mathbb{E}[\|\theta_{k,t+1} - \theta_k^*\|^2] \leq (1 - 2\mu\eta_l)\mathbb{E}[\|\theta_{k,t} - \theta_k^*\|^2] - 2\eta_l(1 - \eta_l L_1)\mathbb{E}[\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)] + \eta_l^2(2\sigma_1^2 + 2\gamma_1^2) + \eta_l\sigma_1^2$$

To account for E local epochs, we model this as E consecutive updates with learning rate η_l. Applying the above inequality E times and using the convexity of $\mathcal{T}(\theta_k)$, we get the following:

$$\mathbb{E}[\|\theta_{k,t}^E - \theta_k^*\|^2] \leq (1 - 2\mu\eta_l)^E\mathbb{E}[\|\theta_{k,t} - \theta_k^*\|^2] - 2E\eta_l(1 - \eta_l L_1)\mathbb{E}[\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)] + E\eta_l^2(2\sigma_1^2 + 2\gamma_1^2) + E\eta_l\sigma_1^2 + 4L_1\eta_l^2E(E-1)\gamma_1^2$$

Finally, using the fact that $\eta_l \leq 1/(4L_1)$ (which follows from our choice of η_l) and combining like terms, we obtain the stated bound:

$$\mathbb{E}[\|\theta_{k,t+1} - \theta_k^*\|^2] \leq (1 - \mu\eta_l)\mathbb{E}[\|\theta_{k,t} - \theta_k^*\|^2] - 2\eta_l\mathbb{E}[\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)] + \eta_l^2(\sigma_1^2 + 6L\gamma E\eta_l + 8(E-1)^2\gamma_1^2) \quad (9)$$

Lemma 2 (One-step Progress for PPGM*):* For any round $t \geq 0$, the convergence bound for *PPGM* follows:

$$\mathbb{E}[\|\psi_{k,t+1} - \psi_k^*\|^2] \leq \mathbb{E}[\|\psi_{k,t} - \psi_k^*\|^2] - 2\eta_g\mathbb{E}[\mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*)] - \frac{\eta_g}{L_2}\sigma_2^2 + \eta_g^2\sigma_2^2 \quad (10)$$

Proof 2: Let $g_t' = \nabla\mathcal{V}(\psi_{k,t})$ be the gradient of the *PPGM* loss function at round t. The update rule for ψ_k with learning rate η_g defined as:

$$\psi_{k,t+1} = \psi_{k,t} - \eta_g g_t' \quad (11)$$

We begin by expanding $\|\psi_{k,t+1} - \psi_k^*\|^2$:

$$\|\psi_{k,t+1} - \psi_k^*\|^2 = \|\psi_{k,t} - \eta_g g_t' - \psi_k^*\|^2 = \|\psi_{k,t} - \psi_k^*\|^2 - 2\eta_g\langle g_t', \psi_{k,t} - \psi_k^* \rangle + \eta_g^2\|g_t'\|^2$$

Taking the expectation of both sides:

$$\mathbb{E}[\|\psi_{k,t+1} - \psi_k^*\|^2] = \mathbb{E}[\|\psi_{k,t} - \psi_k^*\|^2] - 2\eta_g\mathbb{E}[\langle g_t', \psi_{k,t} - \psi_k^* \rangle] + \eta_g^2\mathbb{E}[\|g_t'\|^2] \quad (12)$$

By the L-smoothness of $Y(\rho_k, \varrho_k)$ with Lipschitz constant $L_2 > 0$, we have:

$$\mathcal{V}(\psi_k^*) \geq \mathcal{V}(\psi_{k,t}) + \langle g_t', \psi_k^* - \psi_{k,t} \rangle + \frac{1}{2L_2}\|g_t'\|^2 \quad (13)$$

Rearranging this inequality:

$$\langle g_t', \psi_k^* - \psi_{k,t} \rangle \geq \mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*) + \frac{1}{2L_2}\|g_t'\|^2 \quad (14)$$

Substituting Equation 14 into Equation 12, we have:

$$\mathbb{E}[\|\psi_{k,t+1} - \psi_k^*\|^2] \leq \mathbb{E}[\|\psi_{k,t} - \psi_k^*\|^2] - 2\eta_g\mathbb{E}[\mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*)] - \frac{\eta_g}{L_2}\mathbb{E}[\|g_t'\|^2] + \eta_g^2\mathbb{E}[\|g_t'\|^2]$$

Now, using the assumption of bounded variance of stochastic gradients, $\mathbb{E}[\|g_t\|^2] \leq \sigma_2^2$, we get:

$$\mathbb{E}[\|\psi_{k,t+1} - \psi_k^*\|^2] \leq \mathbb{E}[\|\psi_{k,t} - \psi_k^*\|^2] - 2\eta_g\mathbb{E}[\mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*)] - \frac{\eta_g}{L_2}\sigma_2^2 + \eta_g^2\sigma_2^2$$

Theorem 1 (Convergence of ProtoGen *in Convex Settings):* Let Assumptions 1-4 hold. Let the learning rates be set as $\eta_l = \frac{2}{\mu(t+\gamma_1)}$ for client models and $\eta_g = \frac{2}{L_2(t+\gamma_2)}$ for the *PPGM*, where $\gamma_1 = \max\{\frac{8L_1}{\mu}, E\}$, $\gamma_2 = \max\{\frac{8L_2}{\mu}, E\}$, and E is the number of local epochs. Then, for $T \geq 1$ number of global rounds of training, the output of ProtoGen satisfies, following convergence bound:

$$\mathbb{E}[\mathcal{T}(\bar{\theta}_{k,T})] - \mathcal{T}(\theta_k^*) \leq \frac{4L_1\gamma_1}{\mu T}\left(1 + \log\left(\frac{T}{\gamma_1} + 1\right)\right) \quad (15)$$

$$\mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T})] - \mathcal{V}(\psi_k^*) \leq \frac{4L_2\gamma_2}{T}\left(1 + \log\left(\frac{T}{\gamma_2} + 1\right)\right) \quad (16)$$

where $\bar{\theta}_{k,T} = \frac{1}{T}\sum_{t=1}^T \theta_{k,t}$, $\bar{\psi}_T = \frac{1}{T}\sum_{t=1}^T \psi_t$, and θ_k^*, ψ^* are the respective optimal parameters.

Proof 3 (Theorem 1 (Convex Client Model Bound)): We first derive the convergence bound for the local model of any client k. By Lemma 1, summing over T rounds gives:

$$\sum_{t=0}^{T-1} 2\eta_{l,t}\mathbb{E}[\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)] \leq \|\theta_{k,0} - \theta_k^*\|^2 + \sum_{t=0}^{T-1}\eta_{l,t}^2 C, \quad (17)$$

where $C = \sigma_1^2 + 6L_1\gamma_1 E\eta_{l,t} + 8(E-1)^2\gamma_1^2$ and the learning rate is $\eta_{l,t} = \frac{2}{\mu(t+\gamma_1)}$.

Let $\Delta_t := \mathbb{E}[\mathcal{T}(\theta_{k,t}) - \mathcal{T}(\theta_k^*)]$. Then the inequality becomes:

$$\sum_{t=0}^{T-1} \frac{4}{\mu(t+\gamma_1)} \Delta_t \leq D + \sum_{t=0}^{T-1} \frac{4}{\mu^2(t+\gamma_1)^2} C, \tag{18}$$

where $D = \|\theta_{k,0} - \theta_k^*\|^2$.

Next, apply Jensen's inequality and convexity of $\mathcal{T}(\cdot)$:

$$\mathbb{E}[\mathcal{T}(\bar{\theta}_{k,T}) - \mathcal{T}(\theta_k^*)] \leq \frac{1}{T} \sum_{t=0}^{T-1} \Delta_t. \tag{19}$$

We now bound the harmonic and squared harmonic series using integral approximation:

$$\sum_{t=0}^{T-1} \frac{1}{t+\gamma_1} \leq \log\left(\frac{T}{\gamma_1} + 1\right), \tag{20}$$

$$\sum_{t=0}^{T-1} \frac{1}{(t+\gamma_1)^2} \leq \frac{1}{\gamma_1}. \tag{21}$$

Thus,

$$\sum_{t=0}^{T-1} \Delta_t \leq \frac{\mu D}{4} \log\left(\frac{T}{\gamma_1} + 1\right) + \frac{C}{\mu\gamma_1}. \tag{22}$$

Dividing by T, we get:

$$\mathbb{E}[\mathcal{T}(\bar{\theta}_{k,T}) - \mathcal{T}(\theta_k^*)] \leq \frac{\mu D}{4T} \log\left(\frac{T}{\gamma_1} + 1\right) + \frac{C}{\mu\gamma_1 T}. \tag{23}$$

By absorbing constants into a single constant (e.g., involving L_1), we obtain:

$$\mathbb{E}[\mathcal{T}(\bar{\theta}_{k,T})] - \mathcal{T}(\theta_k^*) \leq \frac{4L_1\gamma_1}{\mu T} \left(1 + \log\left(\frac{T}{\gamma_1} + 1\right)\right), \tag{24}$$

which proves the desired bound for client models.

Proof 4 (Theorem 1 (Convex PPGM *Bound)):* To derive the convergence for the *PPGM* model, we start by summing the result of Lemma 2 over all T rounds:

$$\sum_{t=0}^{T-1} 2\eta_g \mathbb{E}[\mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*)] \leq \mathbb{E}[\|\psi_{k,0} - \psi_k^*\|^2] + T\eta_g^2\sigma_2^2. \tag{25}$$

Using Jensen's inequality and convexity of $\mathcal{V}(\cdot)$:

$$\mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T}) - \mathcal{V}(\psi_k^*)] \leq \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*)]. \tag{26}$$

Multiply both sides of Equation equation 26 by $2T\eta_g$:

$$2T\eta_g \mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T}) - \mathcal{V}(\psi_k^*)] \leq 2\eta_g \sum_{t=0}^{T-1} \mathbb{E}[\mathcal{V}(\psi_{k,t}) - \mathcal{V}(\psi_k^*)]. \tag{27}$$

Applying Equation equation 25, we get:

$$2T\eta_g \mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T}) - \mathcal{V}(\psi_k^*)] \leq \mathbb{E}[\|\psi_{k,0} - \psi_k^*\|^2] + T\eta_g^2\sigma_2^2. \tag{28}$$

Divide both sides by $2T\eta_g$:

$$\mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T}) - \mathcal{V}(\psi_k^*)] \leq \frac{\mathbb{E}[\|\psi_{k,0} - \psi_k^*\|^2]}{2T\eta_g} + \frac{\eta_g\sigma_2^2}{2}. \tag{29}$$

Now use the learning rate $\eta_g = \frac{2}{L_2(t+\gamma_2)}$ with $\gamma_2 = \max\left\{\frac{8L_2}{\mu}, E\right\}$. Since η_g is non-increasing, we have:

$$\eta_g \leq \frac{2}{L_2\gamma_2}. \tag{30}$$

Substituting this into Equation equation 29:

$$\mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T}) - \mathcal{V}(\psi_k^*)] \leq \frac{L_2\gamma_2\mathbb{E}[\|\psi_{k,0} - \psi_k^*\|^2]}{4T} + \frac{\sigma_2^2}{L_2\gamma_2}. \tag{31}$$

Now, applying the L_2-smoothness of $\mathcal{V}(\cdot)$:

$$\mathbb{E}[\|\psi_{k,0} - \psi_k^*\|^2] \leq \frac{2}{L_2}(\mathcal{V}(\psi_{k,0}) - \mathcal{V}(\psi_k^*)). \tag{32}$$

Thus, we get:

$$\mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T}) - \mathcal{V}(\psi_k^*)] \leq \frac{\gamma_2(\mathcal{V}(\psi_{k,0}) - \mathcal{V}(\psi_k^*))}{2T} + \frac{\sigma_2^2}{L_2\gamma_2}. \tag{33}$$

Finally, this bound can be simplified using the inequality $\frac{\gamma_2}{2T} \leq \frac{4\gamma_2}{T}(1+\log(\frac{T}{\gamma_2}+1))$ for $T \geq 1$, and absorbing constants:

$$\mathbb{E}[\mathcal{V}(\bar{\psi}_{k,T})] - \mathcal{V}(\psi_k^*) \leq \frac{4L_2\gamma_2}{T} \left(1 + \log\left(\frac{T}{\gamma_2} + 1\right)\right), \tag{34}$$

Theorem 2 (Non-Convex Convergence Bound for Client Model): Let Assumptions 1–4 hold. Suppose a fixed client k performs local SGD with step size η_l for T rounds, each consisting of E local updates. Then, the iterates $\{\theta_{k,t}\}_{t=0}^{T-1}$ satisfy:

$$\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla\mathcal{T}(\theta_{k,t})\|^2] \leq \frac{2}{\eta_l}(\mathcal{T}(\theta_{k,0}) - \mathcal{T}(\theta_k^*)) + L_1 T\eta_l\sigma_1^2. \tag{35}$$

$$\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla\mathcal{V}(\psi_{k,t})\|^2] \leq \frac{2}{\eta_g}(\mathcal{V}(\psi_{k,0}) - \mathcal{V}(\psi_k^*)) + L_2 T\eta_g\sigma_2^2. \tag{36}$$

Proof 5 (Theorem 2 (Non-Convex Client Model Bound)): The local SGD update rule for client k at round t is:

$$\theta_{k,t+1} = \theta_{k,t} - \eta_l g_{k,t}, \tag{37}$$

where $g_{k,t}$ is the stochastic gradient estimator. Under the L_1-smoothness of $\mathcal{T}(\theta_k)$, we have:

$$\mathcal{T}(\theta_{k,t+1}) \leq \mathcal{T}(\theta_{k,t}) - \eta_l\langle\nabla\mathcal{T}(\theta_{k,t}), g_{k,t}\rangle + \frac{L_1\eta_l^2}{2}\|g_{k,t}\|^2. \tag{38}$$

Taking expectation conditioned on $\theta_{k,t}$ and using $\mathbb{E}[g_{k,t}] = \nabla\mathcal{T}(\theta_{k,t})$:

$$\mathbb{E}[\mathcal{T}(\theta_{k,t+1})] \leq \mathcal{T}(\theta_{k,t}) - \eta_l\|\nabla\mathcal{T}(\theta_{k,t})\|^2 + \frac{L_1\eta_l^2}{2}\mathbb{E}[\|g_{k,t}\|^2]. \tag{39}$$

Using the variance bound $\mathbb{E}[\|g_{k,t}\|^2] \leq \|\nabla\mathcal{T}(\theta_{k,t})\|^2 + \sigma_1^2$, we get:

$$\mathbb{E}[\mathcal{T}(\theta_{k,t+1})] \leq \mathcal{T}(\theta_{k,t}) - \left(\eta_l - \frac{L_1\eta_l^2}{2}\right)\|\nabla\mathcal{T}(\theta_{k,t})\|^2 + \frac{L_1\eta_l^2}{2}\sigma_1^2. \tag{40}$$

Summing over $t = 0$ to $T - 1$, we obtain:

$$\sum_{t=0}^{T-1} \left(\eta_l - \frac{L_1 \eta_l^2}{2} \right) \mathbb{E}[\|\nabla \mathcal{T}(\theta_{k,t})\|^2] \leq \mathcal{T}(\theta_{k,0}) - \mathcal{T}(\theta_k^*) + \frac{L_1 T \eta_l^2}{2} \sigma_1^2. \tag{41}$$

Assuming $\eta_l \leq \frac{1}{L_1}$, we simplify the left-hand side:

$$\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{T}(\theta_{k,t})\|^2] \leq \frac{2}{\eta_l} \left(\mathcal{T}(\theta_{k,0}) - \mathcal{T}(\theta_k^*) \right) + L_1 T \eta_l \sigma_1^2. \tag{42}$$

Proof 6 (Theorem 2 (Non-Convex PPGM Bound)): Identical in structure to Proof 5, replacing $T, \theta, \eta_l, L_1, \sigma_1^2$ with $V, \psi, \eta_g, L_2, \sigma_2^2$.

D. Privacy Analysis of ProtoGen

Let each client maintain a feature extractor $\mathcal{H}(\phi_k, x)$ and a *PPGM* model with encoder $\mathcal{E}(\rho_k, z|y)$ and decoder $\mathcal{J}(\varrho_k, e|y)$. The privacy mechanism relies on input perturbation, followed by feature extraction and training a generator only on the perturbed representations.

Assumption 6 (Input Perturbation by Gaussian Noise): Each input x_i is perturbed using additive Gaussian noise:

$$\tilde{x}_i = x_i + \xi_i, \quad \text{where } \xi_i \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assumption 7 (Lipschitz Continuity of the Feature Extractor): The feature extractor $\mathcal{H}$ is L-Lipschitz continuous, i.e., for all $x, x' \in \mathbb{R}^d$,

$$\|\mathcal{H}(\phi_k, x) - \mathcal{H}(\phi_k, x')\|_2 \leq L \cdot \|x - x'\|_2.$$

Definition 1 (Differential Privacy [1]): A randomized algorithm $\mathcal{M}$ satisfies (ε, δ)-differential privacy if for any two adjacent datasets $\mathcal{D}, \mathcal{D}'$ differing in at most one record, and for any measurable subset S of outputs,

$$\Pr[\mathcal{M}(\mathcal{D}) \in S] \leq e^{\varepsilon} \Pr[\mathcal{M}(\mathcal{D}') \in S] + \delta.$$

Definition 2 (Global Sensitivity): The L_2-sensitivity of a function $\mathcal{H}(\phi_k, \mathcal{X}_k) \to \mathcal{Z}_k$ is

$$\Delta_h = \sup_{\mathcal{D}, \mathcal{D}' \text{ differ in one sample}} \|\mathcal{H}(\phi_k, \mathcal{D}) - \mathcal{H}(\phi_k, \mathcal{D}')\|_2.$$

Theorem 3 (Gaussian Mechanism [1]): Let $\mathcal{H}(\phi_k, \mathcal{X}_k) \to \mathcal{Z}_k$ be a function with L_2 sensitivity Δ_h. Then the mechanism

$$\mathcal{M}(\mathcal{D}) = \mathcal{H}(\phi_k, \mathcal{D}) + \mathcal{N}(0, \sigma^2 I_d)$$

is (ε, δ)-differentially private for any $\delta > 0$, where

$$\varepsilon = \frac{\Delta_f^2}{2\sigma^2} + \log(1/\delta).$$

Lemma 3 (Differential Privacy of Noisy Feature Extraction): Let $\mathcal{M}(x) = \mathcal{H}(\phi_k, x + \xi)$ where $\xi \sim \mathcal{N}(0, \sigma^2 I_d)$, and let the input sensitivity $\Delta_x := \sup_{x,x'} \|x - x'\|_2$ be bounded. Then $\mathcal{M}$ satisfies (ε, δ)-differential privacy with

$$\varepsilon = \frac{L^2 \Delta_x^2}{2\sigma^2} + \log(1/\delta).$$

Proof 7: We consider two neighboring datasets that differ in one input point x_i versus x_i'. We analyze the output of the mechanism $\mathcal{M}(x) = \mathcal{H}(\phi_k, x + \xi)$. Since $\mathcal{H}$ is L-Lipschitz and ξ is sampled independently from $\mathcal{N}(0, \sigma^2 I_d)$:

$$\|\mathcal{M}(x) - \mathcal{M}(x')\|_2 = \|\mathcal{H}(\phi_k, x + \xi) - \mathcal{H}(\phi_k, x' + \xi)\|_2 \tag{43}$$

Using assumption 7, we have following inequality:

$$\|\mathcal{M}(x) - \mathcal{M}(x')\|_2 \leq L \cdot \|x - x'\|_2. \tag{44}$$

Hence, the sensitivity of the mapping $x \mapsto \mathcal{H}(\phi_k, x + \xi)$ is at most $L \cdot \Delta_x$. By applying the Gaussian mechanism, this perturbation ensures (ε, δ)-DP for

$$\varepsilon = \frac{(L\Delta_x)^2}{2\sigma^2} + \log(1/\delta).$$

Lemma 4 (Post-Processing Invariance [1]): Let $\mathcal{M} : \mathcal{D} \to \mathcal{Y}$ be an (ε, δ)-DP mechanism. Let $f : \mathcal{Y} \to \mathcal{Z}$ be any (possibly randomized) function. Then $f \circ \mathcal{M}$ also satisfies (ε, δ)-DP.

Theorem 4 (Privacy Guarantee of PPGM *Training and Sharing):* Let each feature z_i used for *PPGM* model training be generated via $z_i = \mathcal{H}(\phi_k, x_i + \xi_i)$, where $\xi_i \sim \mathcal{N}(0, \sigma^2 I_d)$ and $\mathcal{H}$ is L-Lipschitz. Then:

1) The feature memory $\tilde{\mathcal{D}}_k = \{(\tilde{z}_i, y_i)\}_{i=1}^m$ satisfies (ε, δ)-differential privacy with respect to the original data $\mathbb{D}_k$.
2) The decoder $\mathcal{J}_{\varrho_k}$ trained on $\tilde{\mathcal{D}}_k$ also satisfies (ε, δ)-DP due to post-processing.

Proof 8 (Theorem 4):

Step 1: Privacy of Feature Extraction. From Lemma 3, each feature $\tilde{z}_i = \mathcal{H}(\phi_k, x_i + \xi_i)$ is (ε, δ)-DP due to the Gaussian mechanism applied to the input x_i, and the Lipschitz property of $\mathcal{H}$.

Step 2: Privacy of *PPGM* model Training. The *PPGM* is trained on the dataset $\tilde{\mathcal{D}}_k$ composed of $(\tilde{z}_i, y_i)$. Since the entire training procedure (gradient descent) operates on already private representations, and no raw x_i or unperturbed z_i are used, by Lemma 4 (post-processing), the resulting trained model — including the decoder $\mathcal{J}(\varrho_k, e)$ — is also (ε, δ)-DP.

Step 3: Sharing the Decoder. Only the decoder parameters ϱ_k are transmitted to the server. These parameters cannot be reversed for any particular $\tilde{z}_i$ due to aggregated training and noise injection. Because $\mathcal{J}(\varrho_k, e)$ is computed via post-processing of DP data, it retains the same (ε, δ) privacy guarantees.

Thus, the privacy of any individual data sample in the client dataset $\mathbb{D}_k$ is preserved throughout the entire learning procedure.

REFERENCES

[1] C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*. Foundations and Trends in Theoretical Computer Science, now Publishers Inc., 2014.